SRIVARMA BATTINI

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About Me

Motivated B.Tech CSE (AI) student with strong expertise in Machine Learning, Deep Learning, and NLP. Experienced in building end-to-end AI pipelines spanning classical ML models, deep neural networks and large language models. Proficient in developing RAG systems, fine-tuning transformer models, and deploying scalable AI solutions using FastAPI and cloud-ready infrastructure.

Education

Amrita Vishwa Vidyapeetham, Amritapuri Campus — **Aug 2023 – Present**
Bachelor of Technology in Computer Science and Engineering (AI) — GPA: 8.91

Sri Chaitanya Junior College — **2021 – 2023**
Higher Secondary Education (Class XII) — Percentage: 97.4%

Jawahar Navodaya Vidyalaya, Srikakulam — **2016 – 2021**
Secondary School (Class X) — Percentage: 89.6%

Technical Skills

Backend & Programming: Python, FastAPI, REST API, Git
ML & DL: Regression, Classification, CNNs, RNNs, LSTMs, Transformers, BERT, Computer Vision, Scikit-learn, PyTorch
Generative AI & LLMs: RAG, SBERT, Llama 3.1, Groq API, ChromaDB, Prompt Engineering
Databases: SQL, PostgreSQL, MongoDB, PL/SQL
Big Data Technologies: Hadoop, Apache Spark, HDFS, MapReduce
Cloud Technologies: Docker, Kubernetes, Cloud Computing
Core CS: Data Structures & Algorithms, Probability, Linear Algebra, Statistics

Projects

EmailIQ: Controlled Email Tone and Politeness Optimization — **Mar 2026 – May 2026**
Tech: DeBERTa, RAG, SBERT, ChromaDB, Llama 3.1, Groq API, Python

- Built an NLP system using **7 fine-tuned DeBERTa models** to classify tone, intent, politeness, sarcasm, passive aggression, urgency, and power dynamics in workplace emails.
- Designed a **RAG pipeline** with SBERT embeddings and ChromaDB, integrating Llama 3.1 via Groq API to generate professional rewrites while preserving original intent.
- Engineered risk assessment and professionalism scoring modules for actionable workplace communication feedback.

Constraint-Based Intelligent Travel Planning System — **Oct 2025 – Jan 2026**
Tech: React, FastAPI, REST API, MongoDB, Python, Dynamic Programming, TSP

- Built a full-stack app (React + FastAPI + MongoDB) generating personalized itineraries from user constraints (budget, duration, interests).
- Implemented **Dynamic Programming** for K-best itinerary generation, **0/1 Knapsack** for place selection, and **TSP** (Nearest Neighbour + 2-opt) for route optimization.

Brain Tumor MRI Segmentation & Survival Prediction — **Jul 2025 – Aug 2025**
Tech: PyTorch, ResNet, U-Net, NIfTI, Scikit-learn, Python

- Developed a deep learning segmentation pipeline using **ResNet-based U-Net** on 3D MRI NIfTI scans with slice-wise preprocessing and Dice loss for class imbalance handling.
- Extracted tumor morphological features from segmentation outputs for downstream **patient survival regression modeling**.

Competitive Programming Repository

- Maintained a structured repository of DSA and competitive programming solutions in Python covering graphs, dynamic programming, trees, and sorting with complexity analysis.

Certificates

- Google Cloud Computing Foundations Certificate – Google Cloud
- Elements of AI – MinnaLearn & University of Helsinki
- Deep Learning with PyTorch – DataCamp